FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

FULL TEST – XI

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*
   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.
   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*
2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*
3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):
(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

Physics

PART – I

SECTION – A

Straight Objective Type

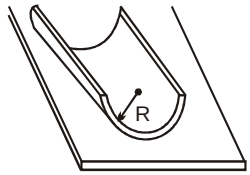
This section contains **3 multiple choice questions**. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

1. If a loop of chain is spun at high speed, it will roll like a loop without collapsing. Consider a chain of linear mass density 'μ' that is rolling without slipping at a high speed v_0. The chain rolls over a small bump and a transverse wave pulse is generated in the chain. How far around the loop (in radians) will a transverse wave pulse travel in the time the loop rolls through one complete revolution.

(A) π (B) 2π

(C) 3π (D) 4π

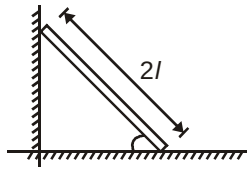
2. The semicircular cylindrical shell of radius R with small but uniform wall thickness is set into small oscillation on the horizontal surface. If no slipping occurs, then the expression for the period T of each complete oscillation is:



(A) $T = 2\pi\sqrt{\frac{(\pi-2)R}{g}}$ (B) $T = 2\pi\sqrt{\frac{(\pi-4)R}{g}}$

(C) $T = 2\pi\sqrt{\frac{2R}{g}}$ (D) $T = 2\pi\sqrt{\frac{(\pi-6)R}{g}}$

3. A plank of length 2L leans against a vertical wall. It starts to slip downward without friction. The top of the plank loses contact with the wall when it is at a height of



(A) $\frac{3}{4}^{th}$ of its initial height

(B) half of its initial height

(C) $\frac{1}{4}^{th}$ of its initial height

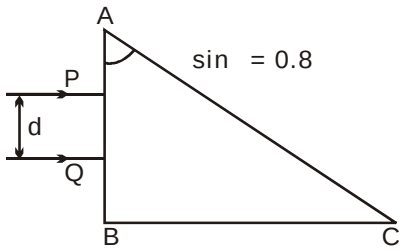
(D) $\frac{2}{3}^{rd}$ of its initial height

Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

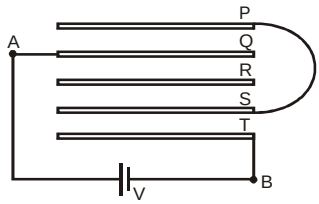
4. Two parallel beams of light P and Q (separation d) containing radiation of wavelength 4000 Å and 5000 Å are incident normally on a prism as shown in figure. The refractive index of the prism as a function of wavelength is given by $\mu(\lambda) = 1.20 + \frac{b}{\lambda^2}$ where λ (in Å) and b is a positive constant. The value of b is such that the condition for total internal reflection of the face AC is satisfied for one wavelength and not for the other then $[\sin 80.3 = 0.9856]$



(A) The value of $b = 8 \times 10^{-15}\ m^2$.
(B) Deviation of beam transmitted through face AC is 27.2^0.
(C) The value of $b = 4 \times 10^{-15}\ m^2$
(D) Deviation of beam transmitted through face AC is 38.7^0

5. In a photoelectric effect set up, a point source of light of power 3.2×10^{-3} W emits monoenergetic photons of energy 5.0 eV. The source is located at a distance of 0.8m from the centre of a stationary metallic sphere of work function 3.0 eV and of radius 8×10^{-3}m. The efficiency of photo electrons emission is one for every 10^6 incident photons. Assume sphere is isolated and initially neutral and that photo-electrons are instantly swept away after emission then.
(A) No. of electrons emitted per second is 10^6
(B) Time after which electron emission stops is 111.11 sec
(C) No. of electrons emitted per second is 10^5.
(D) Time after which electron emission stops is 11.11 sec

6. Five plates are arranged as shown in figure and connected across a battery of emf V. The separation between each plate is 'd' and the surface area of each plates is 'A'.



(A) Equivalent capacity between A and B is $\frac{\varepsilon_0 A}{5d}$
(B) Equivalent capacity between A and B is $\frac{3\varepsilon_0 A}{5d}$
(C) Charge on plates P is $\frac{\varepsilon_0 AV}{5d}$
(D) Charge on plate R is $\frac{3\varepsilon_0 AV}{5d}$

Space for rough work

7. Which of the following statement(s) is/are correct:-
   (A) If the electric field due to a point charge varies as $r^{-2.5}$ instead of r^{-2}, then Gauss law will still be valid.
   (B) The Gauss law can be used to calculate the field distribution around an electric dipole
   (C) If the field between the two point charges is zero somewhere, then the sign of charge is same.
   (D) The work done by external force in moving a unit positive charge from point A at potential V_A to point B at potential V_B is $V_B - V_A$.

8. An external magnetic field is decreased to zero, due to which a current is induced in a small circular wire loop of radius 'r' and resistance 'R' placed in the field. This current will not become zero,
   (A) At the instant when external magnetic field stops changing (t = 0), the current in the loop as a function of time for t > 0 is given by $i = i_0\ e^{-2RT/\mu_0\pi r}$
   (B) At the instant B stops changing (t = 0) the current in the loop as a function of time t > 0 is given by $i = \dfrac{\mu_0 i_0 R}{2r}$
   (C) The time in which current in loop decreases to $10^{-3} I_0$ (from t = 0) for R = 100Ω and r = 5 cm is given by $\dfrac{3\pi^2 \ln 10}{10^{10}}$ sec.
   (D) The time in which current in loop decreases to $10^{-3} I_0$ (from t = 0) for $R = 100\Omega$ and r = 5 cm is given by $\dfrac{3\pi^2}{10^6}$ sec.

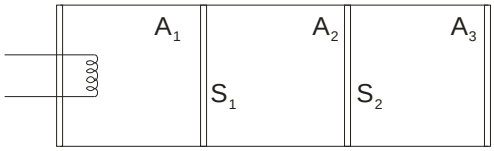
Space for rough work

Comprehension Type

This section contains **1 group of questions.** The group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 9 and 10

Consider a closed cylinder whose walls are adiabatic. The cylinder lies in a horizontal position and is divided into three parts A_1, A_2, A_3 by mean of a partition S_1 and S_2 which can move along the length of the cylinder without friction. Piston S_1 is adiabatic while piston S_2 is conducting. Initially each part contains one mole of Helium gas at temperature T_0, pressure P_0, volume v_0 (see figure). Now heat is slowly supplied to the gas in part A_1 till the temperature in part A_3 becomes $T_3 = \frac{9T_0}{4}$.



9. The final equilibrium temperature of A_1 is:

(A) $\frac{9T_0}{4}$ (B) $\frac{585}{24}T_0$

(C) $\frac{585T_0}{32}$ (D) $\frac{585T_0}{36}$

10. The final volume of A_1 is:

(A) $\frac{8V_0}{27}$ (B) $\frac{65V_0}{18}$

(C) $\frac{65V_0}{27}$ (D) $\frac{65V_0}{9}$

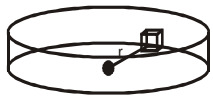
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SECTION – C

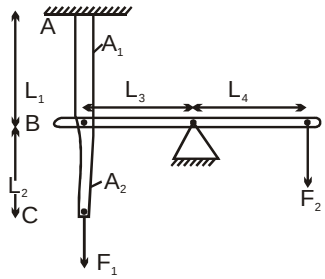
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

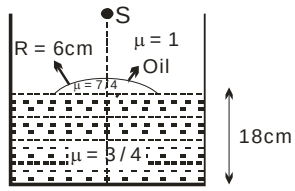
11. A block of mass 'm' slides on a circular track of radius 'r' whose wall and floor both have coefficient of kinetic friction ' μ ' with the block. The size of block is small compared to the radius of the track. The floor lies in horizontal plane and the wall is vertical. The block is in contact with both wall and the floor. The block has initial speed υ_0. For the block to circle the track exactly once before coming to rest, v_0 is given by $\sqrt{gr(e^{K\pi\mu}-1)}$. Find the value of K.



12. The assembly shown in figure is loaded by forces F_1 and F_2. Assuming that both parts of the vertical bar ABC are made of same material, if net vertical displacement of point C is zero then ratio of $\frac{F_2}{F_1}$ is:
Given: $\left[A_1 = 2A_2,\ L_3 = L_4,\ L_1 = 2L_2\right]$.

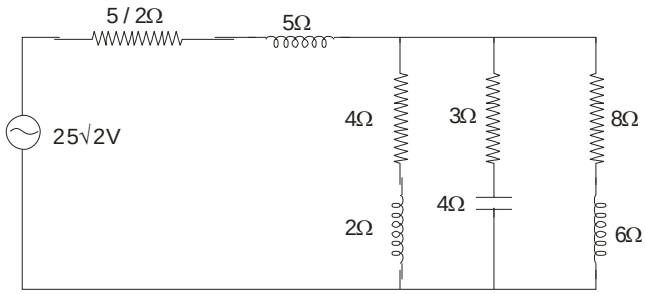


13. Water (with refractive index $\mu = \frac{4}{3}$) in a tank is 18 m deep. Oil of refractive index $\frac{7}{4}$ lies on water making a convex surface of radius of curvature R = 6 cm as shown. Consider oil to act as a thin lens. An object is placed 24 cm above water surface. The location of its image is at *x* cm above the bottom of the tank. Then find value of *x*.

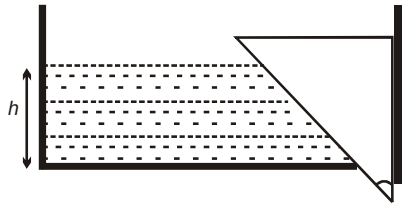


Space for rough work

14. Consider a network shown in figure. The current supplied by the AC voltage source is n amperes. What is the value of n.



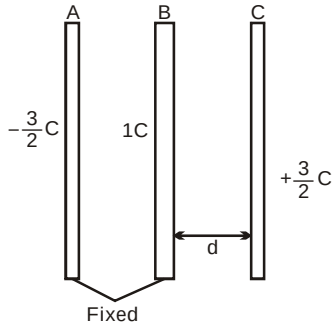
15. A slit is cut along the right bottom edge of a rectangular tank. The slit is closed by a wooden wedge of mass m and apex angle θ as shown in figure. The vertical plane surface of the wedge is in contact with the right vertical wall of the tank. Coefficient of static friction between these surfaces in contact is μ. If the maximum height upto which the water can be filled in the tank without any leakage is:



Given: $\sqrt{\dfrac{Km}{\delta b \tan\theta - \mu}}$. Find the value of K.

where $\rightarrow$ $\delta =$ density of water, b = width of tank.

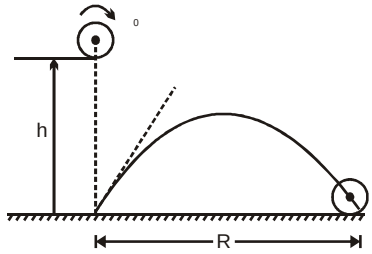
16. Two identical conducting plates A and B with charges $-Q\left(-\dfrac{3}{2}C\right)$ and $+q(1C)$ respectively are located parallel to each other at a small distance. Another identical plate C with mass m and charge $+Q\left(\dfrac{3}{2}C\right)$ is situated parallel to the original plates at a distance d from plate B. (see figure). Area of each plates is S. The plate C is released and can move freely, collides elastically with plate B. [Neglect gravity and boundary effects]. Assume that the charge has enough time to redistribute. The velocity v of plate C after the collision at a distance d from the plate B is $K\sqrt{\dfrac{d}{m\varepsilon_0 S}}$. Find value of K.



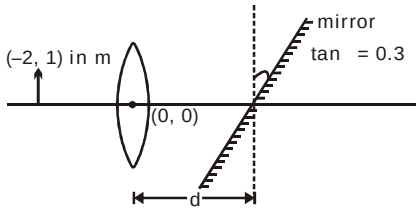
Space for rough work

17. A rocket is launched from a pole of the Earth with velocity V_0, where V_0 is the orbital velocity of rotation near the surface of the Earth, in such a way that it lands at the equator. The rocket's time of flight is T = $\left(\pi+\sqrt{K}\right)\sqrt{\frac{R}{g}}$. Where R $\rightarrow$ Radius of earth, g $\rightarrow$ acc. due to gravity at the surface.
Find value of K.

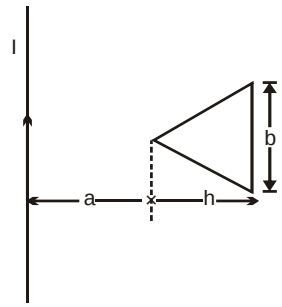
18. Figure shows a solid homogeneous ball of radius R. Before falling, the velocity of centre of mass is zero, but the ball is rotating with angular velocity ω_0 about horizontal axis passing through the centre. The ball is release, it collides with the floor (coefficient of restitution is 'e'). The coefficient of friction between the ball and the floor is μ_k due to which the ball moves in a projectile. Assume that there is slipping throughout the impact and neglect air resistance. The horizontal range of the ball is given by $R = N\mu_k(1+e)eh$. What is the value of N.



19. A convex lens of focal length 1.5 m is placed in a system of co-ordinate axis such that its optical centre is at origin and principle axis coincide with the *x*-axis. An object and a plane mirror are arranged on the principal axis as shown in figure. Find the value of d(in m) so that *y*-co-ordinate of image after refraction and reflection is 0.3 m.



20. A very long straight conductor and an isosceles triangular conductor lie in a plane and are separated from each other as shown in figure. If a = 10 cm, b = 20 cm and h = 10 cm, the coefficient of mutual inductance is given by $N\times10^{-8}[1-\ln 2]H$. Find value of N.



Space for rough work

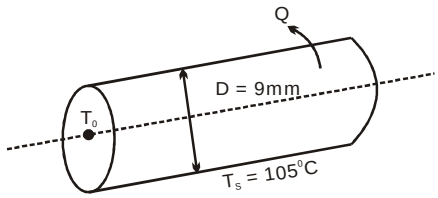
SECTION – D

(Numerical Answer Type)

This section contains **3 questions**. Each question, when worked out will result in **numerical answer type** (XXXXX.XX)

21. Radiation of wavelength 180 nm ejects photoelectrons from a plate whose work function is 2.0 eV. If a magnetic field of flux density 5.0×10^{-5}T is applied parallel to the plate, what would be the radius of path in metres followed by electrons ejected normally from the plate with maximum energy?
($e = -1.6\times10^{-19}$C, $m_e = 9.1\times10^{-31}$kg, $h = 6.6\times10^{-34}$Js, $c = 3\times10^{8}$ m/s)

22. A 2kW resistance heater wire whose thermal conductivity is K = 15 w/m.k. has a diameter of D = 4 mm and a length of L = 0.5 m, and is used to boil water. (see fig). If the outer surface temperature of the resistance wire is T_s = 105 ^{0}C, determine the temperature in ^{0}C at the centre of the wire. [Assume radial heat flow]



23. Consider a following nuclear reaction
$${}^2_1D + {}^2_1D \longrightarrow {}^3_1T + {}^1_1P + \Delta E$$
Given $m\left({}^2_1D\right) = 2.01458\,\text{amu}$, $m\left({}^3_1D\right) = 3.01605\,\text{amu}$, $m\left({}^1_1D\right) = 1.00728\,\text{amu}$.
Calculate the mass in Kg of deuterium $\left({}^2_1D\right)$ required per day for a power output of 10^9W.
Assume that the efficiency of powers to be 50%.

Space for rough work

Chemistry

PART – II

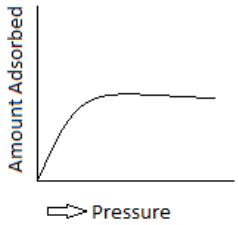
SECTION – A

Straight Objective Type

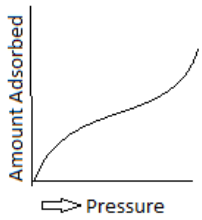
This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

24. 200 gm of anhydrous $FeSO_4$, when dissolved in excess of water produced 84 KJ/mole of heat. Same amount $FeSO_4.7H_2O$, on dissolving in large excess of water absorbs 9.20 KJ/mole. Heat of Hydrogenation of $FeSO_4$ is approx.
(A) 117 KJ (B) 104 KJ
(C) 120 KJ (D) 123.76 KJ

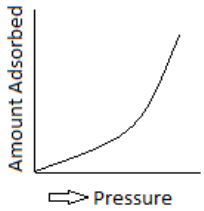
25. Which physical adsorption isotherm graph represents the pressure at which formation of monomolecular layer is complete and that of the multi layer is being started.
(A)



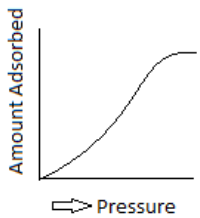
(B)



(C)

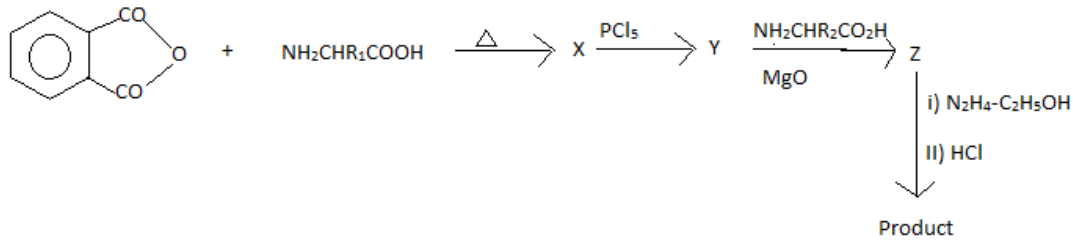


(D)

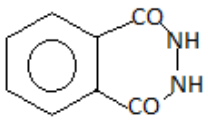


Space for rough work

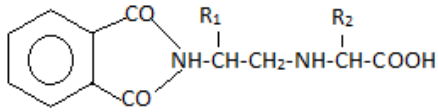
26. Final product for the given reaction below is



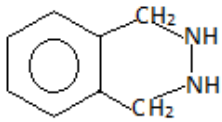
(A)



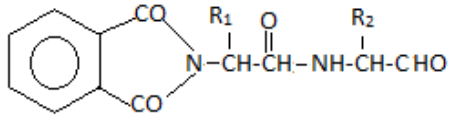
(B)



(C)



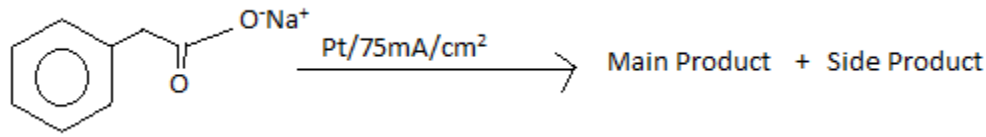
(D)



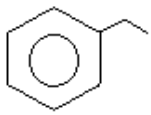
Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

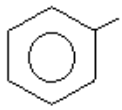
27. Side Product obtained during the electrolysis of given reactant are:



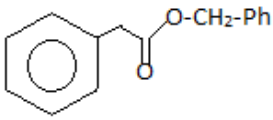
(A)



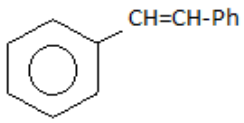
(B)



(C)

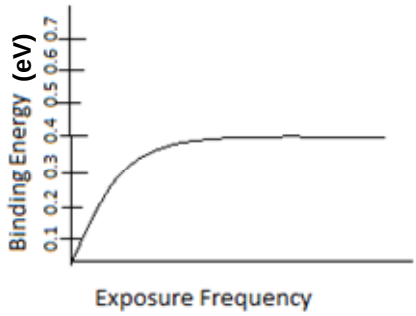


(D)



Space for rough work

28. An electron is ejected from Li^{2+} on exposing 600 nm, the stopping potential & Wavelength of ejected electron is:
(A) 1.675eV, 3.83 x 10^{-10}nm
(B) 1.675eV, 1.23 nm
(C) 2.68 x 10^{-19}J, 1.23 nm
(D) 6.4 x 10^{-20}J, 3.83 x 10^{-10} nm



29. Select the correct option:
(A) $PH_3 > NH_3 > I_3^- > MnO_4^-$ (% p- character)
(B) $PH_3 > NH_3 > I_3^- > MnO_4^-$ (Bond Angle)
(C) $PF_3 > PCl_3 > PBr_3 > PH_3 > PI_3$ (Dipole moment)
(D) $H_2O_2 > H_2O > CH_3COCH_3 > HF$ (Boiling Point)

30. An unknown inorganic salt give white ppt. X with compound Y which form buffer in excess weak base. When heated with weak base in alkaline medium give compound which is the main component to distinguish aldehyde & ketone. Compound X & Y are:
(A) X = $Fe(NO_3)_2$; Y= $Fe(OH)_3$
(B) X= AgCl; Y=$[Ag(NH_3)_2]^+$
(C) X = AgBr; Y=$[Ag(NH_3)_2]^+$
(D) X = $Ag_2O + NH_4Cl$ Y= $AgNO_3 + NH_3 + OH^-$

31. In Fe_3O_4, Fe^{2+} occupy octahedral site because of
(A) High Δ° CFSE (B) Low Δ° CFSE
(C) High Δ_t CFSE (D) Low Δ_t CFSE

Space for rough work

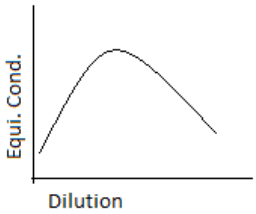
Comprehension Type

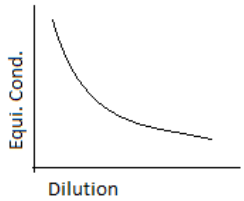
This section contains **1 group of questions.** The group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

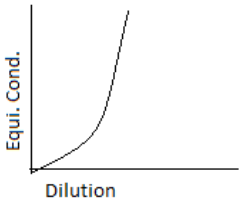
Paragraph for Question Nos. 32 and 33

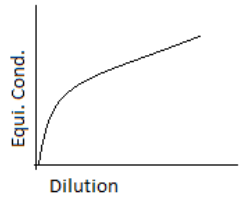
Equivalent point is measured by conductometric titration. We measure the volume of base used to neutralize the acid ions completely in the solution. During the titration, one of the ion is replaced by the other & invariably there two ions differ in the ionic conductivity with the result that conductivity of the solution varies during the course of titration.

32. Which of the following graph represents the correct variation of equivalent conductance with dilution for strong electrolyte?

(A) 

(B) 

(C) 

(D) 

33. The minimum Weight of NaOH required to be added in R.H.S. to consume all the H^+ present in R.H.S. of the cell of e.m.f. 0.701V at 25°C before it use
$Zn|Zn^{2+}(0.1M)$ || HCl (1 L) | H_2(1atm.)| Pt
(A) 2.53 gm (B) 1.26 gm
(C) 0.632 gm (D) None

Space for rough work

SECTION – C

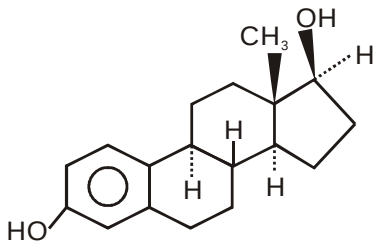
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

34. Number of Species with paramagnetic nature
B_2, NO^+, CO^+, C_2^{2-}, KO_2, Na-NH_3 (dil.)

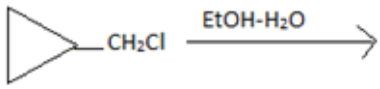
35. Molar mass of gas is 48. If the molar heat of gas at constant volume is 28.9. The atomicity of gas is.

36. No. of stereogenic centre in female sex hormone estradiol is



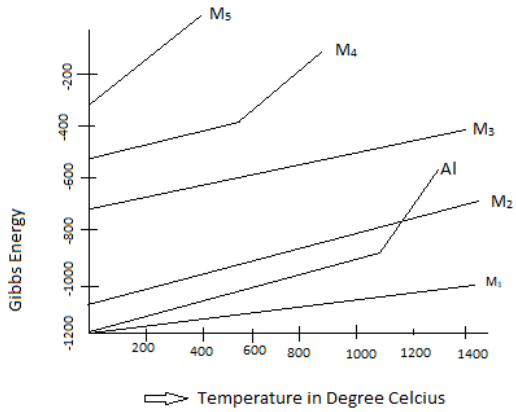
37. In cannizaro reaction, in the presence of 40% base, the order of reaction with respect to aldehyde is

38. Number of possible product in the given reaction below is



Space for rough work

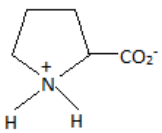
39. In the given Ellingham diagram, the number of species which can be reduced by aluminum above 800 ˚C.

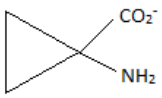


40. During the discharge of lead storage battery, the density of 64% H_2SO_4 (by weight) falls from 1.225 to 0.98 which is 20% (by weight). What is the change in Molarity of H_2SO_4

41. 100 ml of O_2 gas was mixed with 20 ml of a mixture of CH_4 and a gaseous compound of Ethyne & exploded in a eudiometric tube. On cooling at room temperature & pressure, 20 ml volume was observed. After treatment with caustic potash solution, a further contraction of 40 ml was observed. The value of X + Y for the alkyne C_xH_y is

42. Number of correct statements is:
a) Sucrose is a carbohydrate which gives silver mirror test.
b) Starch on complete hydrolysis gives β-D Glucose.
c) Protein gives purple color when treated with ninhydrin
d) Given compounds are α- amino acid





e) Nylon & Polythene are example of step-growth polymer.
f) $TiCl_4 + (CH_3)_3Al$ combination act as catalyst in manufacture of stereo-specific polypropylene.
g) Thermosetting polymer has strongest inter-particle force.
h) Amino glycosides are usually used as antibiotic
i) Parathion is an organic phosphorus compound
j) Butylated Hydroxyanisole (BHA) is used as antioxidant

Space for rough work

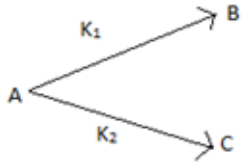
43. The minimum pH required to prevent the precipitation of ZnS in a solution that is 0.01M $ZnCl_2$ & saturated with 5 x 10^{-5} M H_2S. (Given: $K_{sp}(ZnS) = 10^{-17}$; $K(H_2S) = 10^{-21}$)

SECTION – D

(Numerical Answer Type)

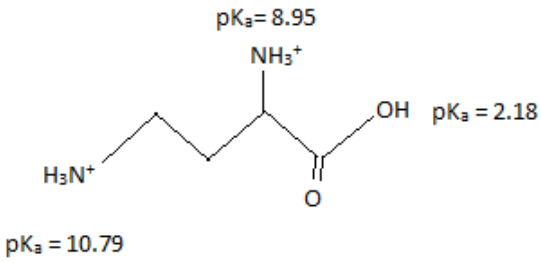
This section contains **3 questions**. Each question, when worked out will result in **numerical answer type** (XXXXX.XX)

44. Consider the reaction
The rate constant for two parallel reaction were found to be 10^{-2} dm^3/mol/sec for B and 3 x 10^{-2} dm^3/mol/sec for C. If the corresponding energy of activation of the parallel reactions are 70 & 110 KJ respectively. What is the apparent net energy of activation in KJ of given reaction?



45. When dry air is passed through 0.03 M solution of $MgCl_2$ & then through 0.07 M glucose solution, there is no loss in mass of glucose solution. Hence the degree of dissociation of $MgCl_2$ is

46. What is the isoelectric point of the following amino acid?



Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **3 multiple choice questions** numbered. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

47. If $2\cos^4 x - 2\cos^3 x + \cos x = \sin^2 x$, then $\sin^2 x =$

(A) 1 (B) $1/2$

(C) $3/4$ (D) $1/4$

48. $p(x) = \int \frac{x^2 \sin 2x - \sin 2x(x\tan x - 1)}{(x^2+1)^2}\, dx$, then $p(1) - p(0) =$

(A) $\frac{1-\cos 2}{4}$ (B) $\frac{-1+\cos 2}{4}$

(C) $1 - \frac{\cos 2}{4}$ (D) None of these

49. $\frac{dy}{dx} = \frac{x}{y} + f\left(\frac{x+y}{y}\right)$ where $y = x.\ln kx;\ k > 0,\ x > 0$, then $f(x) =$

(A) $\frac{3x - x^2 + 1}{x-1}$ (B) $\frac{x^2 - 3x + 1}{1-x}$

(C) $\frac{x^2 + 3x - 1}{x-1}$ (D) $\frac{-x^2 + 3x - 1}{1-x}$

Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

50. Locus of point of intersection of tangents at the ends of a chord subtending an angle of θ, at origin, such that $\tan\theta = a$ for a parabola $y^2 = 4ax$ is:

(A) Parabola if $a = 2$ (B) Hyperbola if $a = 1$

(C) Parabola if $a = 2(1+\sqrt{2})$ (D) Pair of straight line if $a = 1/2$

51. If a_n is no. of ways to split positive integer 'n' in all possible ways, then

(A) $a_n = 2^n$ (B) $a_n = a_{n-1} + a_{n-2};\ n > 2$

(C) $a_n = a_{n-1} + 2 \cdot a_{n-2};\ n > 2$ (D) $a_n = 2^{n-1}$

52. If Z moves on $\arg\left(\frac{Z-2i}{Z-(1+i)}\right) = \pi/3$, then

(A) $\left(\frac{1}{5}+i\right)$ is interior point (B) $\left(\frac{1}{5}+2i\right)$ is interior point

(C) $\left(\frac{1}{5}+i\right)$ is exterior point (D) $\left(\frac{1}{5}+2i\right)$ is exterior point

53. Maximum and minimum value of $\frac{6|Z|^2 - Z^2 - (\bar{Z})^2 - 8}{8}$ are m_1 and m_2 respectively, if Z moves on $Z\bar{Z} - \sqrt{2}\bar{Z}i + \sqrt{2}iZ = 0$, then

(A) $m_1 + m_2 = 6$ (B) $m_1 m_2 = 10$

(C) $\left|\left[\frac{m_1}{m_2}\right]\right| = 7$ (D) $\left[\left|\frac{m_2}{m_1}\right|\right] = 0$

where [.] denotes G.I.F. and |.|is modulus value function.

54. A parabola touches *x*-axis at origin such that foot of perpendicular from origin to axis of parabola is $N(-2, 2)$ then

(A) eqn. of directrix is $x + y + 4 = 0$ (B) Focus of parabola is $(-2, 2)$

(C) Length of latus rectum is $2\sqrt{2}$ (D) eqn. of axis of parabola is $y = x + 4$

Space for rough work

Comprehension Type

This section contains **1 group of questions.** The group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 55 and 56

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + ...\infty$$

55. $\int e^{-\sin\theta} \cos(\cos\theta)\, d\theta =$

(A) $K_1 + \theta + \frac{\cos\theta}{1} - \frac{\sin 2\theta}{2\times 2} - \frac{\cos 3\theta}{3\times 3\times 2} + \frac{\sin 4\theta}{4\times 4\times 6} + ...$

(B) $K_2 - \theta + \frac{\cos\theta}{1} - \frac{\sin 2\theta}{2^2} + \frac{\cos 3\theta}{3^2} + \frac{\sin 4\theta}{4^2} + ...$

(C) $K_3 + \theta - \frac{\cos\theta}{1} + \frac{\sin 2\theta}{2^2} - \frac{\cos 3\theta}{3^2} + \frac{\sin 4\theta}{4^2} + ...$

(D) $K_4 + \theta - \frac{\cos\theta}{1\times 1} - \frac{\sin 2\theta}{2\times 2} - \frac{\cos 3\theta}{3\times 3\times 4} - \frac{\sin 4\theta}{4\times 4\times 4\times 5} + ...$

where K_1, K_2, K_3, K_4 are real constants.

56. $\int e^{\cos\theta}.\sin(\sin\theta)\, d\theta =$

(A) $K_1 - \sin\theta - \frac{\sin 2\theta}{2\times 2!} - \frac{\sin 3\theta}{3\times 3!} - \frac{\sin 4\theta}{4\times 4!} - ....; k_1 \in R$

(B) $K_2 - \cos\theta - \frac{\cos 2\theta}{2\times 2!} - \frac{\cos 3\theta}{3\times 3!} - \frac{\cos 4\theta}{4\times 4!} - ....; k_2 \in R$

(C) $K_3 + \sin\theta + \frac{\sin 2\theta}{2\times 2!} + \frac{\sin 3\theta}{3\times 3!} + \frac{\sin 4\theta}{4\times 4!} + ...; k_3 \in R$

(D) $K_4 + \cos\theta - \frac{\cos 2\theta}{2\times 2!} - \frac{\cos 3\theta}{3\times 3!} + \frac{\cos 4\theta}{4\times 4!} - ....; k_4 \in R$

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

57. If the line $y = x$ intersects the curve $y = x^2 - x + x^{-1}$ at the points P, Q, R and O is the origin then [|OP . OQ . OR |] is _____; where |.| is modulus value function and [.] is G.I.F.

58. $y = \frac{\left[e^x\right]}{x}; x > 0$ has range $[b, \infty]$, then [b] is_____. where [.] is G.I.F.

59. Point P is taken uniquely on lines $x = \sqrt{3}$ and $y = \sqrt{3}(x + \sqrt{3})$ such that angle bisector of $\angle PQR$ in ΔPQR is $y = (\sqrt{3} - 2)x + 2$. If side QR always passes there M (h, k) where PM is not perpendicular to the given angle bisector, then $8h^2 =$

60. If $\cot^4 p + \cot^4 q - 3 = \sin^2(p + q) - 2\tan^2 p \tan^2 q$ where $p, q \in \left(0, \pi/2\right)$ then $[p + q] = ?$

61. On circle $|z - 2i| = 2$; through origin 'O', a chord OP is drawn. On the tangent at O, a point T is taken so that OT = OP. If TP produced meet normal at O to the circle at Q, then limiting value of OQ as P moves towards origin is.

62. No. of ways to put 10 identical balls in 4 identical boxes, so that none of the boxes are empty.

63. If $2x - y - 5 + \lambda(x - 2y - 1) = 0$ represents incident rays and $(3x - y + 4) + \mu(x - 2y + 3)$ represent reflected rays to an ellipse S = 0; where $\lambda, \mu \in \mathbf{R}$. Point P(2, 4) is a point on S = 0, normal at which meets major axis at N. If N divides line segment joining two focii in the ratio a : b (a, b are co-primes) then $a^2 + b^2 - 15$ is.

Space for rough work

64. A line $3x-4y+K=0$ intersects the circle $x^2+y^2-8x-6y+21=0$ at P and Q. If the tangents at P and Q to the circle intersect at $\pi/3$, then |K| is; where |.| is modulus value function.

65. For $x \geq 0,\ F(x)=\int_0^x f(t)\,dt$ and $f(x)$ is a non-negative continuous function. If for some $c>0,\ (cF(x) \geq f(x))$ then, for $x \geq 0,\ f(2)$ is.

66. ΔABC is an acute angle triangle with $\angle B = 30^0$, H is the orthocenter and M is the mid point of BC. On the line HM a point T is taken such that HM = TM. If AT = 8 cm then length of AC is.

SECTION – D

(Numerical Answer Type)

This section contains **3 questions**. Each question, when worked out will result in **numerical answer type** (XXXXX.XX)

67. If $\lambda = \lim\limits_{n\to\infty}\left(\dfrac{1^{18}+2^{18}+3^{18}+.....+n^{18}}{n^{19}}-\dfrac{1^{19}+2^{19}+3^{19}+....+n^{19}}{n^{20}}\right)$; then 1000λ is

68. If $P(x)=\int \dfrac{5x^{-1}+3x^{-3}}{(x^5+2x^{-5})+(2x^3+2x^{-2})+x+2}dx$ then $\tan(P(1)-P(0))$ is.

69. If $x^3dy=(3x^2y+x^2+x^7)\,dx$ with $y(0)=0$ then $y(3)$ is.

Space for rough work